11.13 The Snowplough Problem

WORKSHEET

The question

STEP III Specimen paper, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q11

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The first plough meets snow of depth $h + kt$ at time t. What is its speed? Is it easier to find x as a function of t, or t as a function of x?

STAGE 1 The first plough

Covers: Part (i)

Ask yourself At time t the first plough meets snow of depth $h + kt$. Write its speed. It is easier to find t as a function of x: turn the equation upside down and solve.

STAGE 2 The second plough's equation

Covers: Part (ii), setting up the equation

Pause and try The second plough is at y at time t. The first plough cleared that spot at time $t_1(y)$. How deep is the snow there now? Write the second plough's speed, then turn it upside down.

STAGE 3 Solving

Covers: Part (ii), solving it

Ask yourself The equation is linear in t as a function of y. Use the integrating factor $e^{-\alpha y}$, and the condition that the second plough starts at $y = 0$ at time T.

STAGE 4 The collision

Covers: Part (ii), where the ploughs meet

Ask yourself The ploughs meet when they reach the same point at the same time. Find $t_2(y) - t_1(y)$. When is it zero?

What's really going on

Agnew's snowplough

In Ralph Palmer Agnew's version, from his 1942 textbook *Differential Equations*, it starts snowing heavily and steadily one morning; a snowplough sets out at noon and covers 2 miles in the first hour and 1 mile in the second; when did it start snowing? The plough's speed is inversely proportional to the depth, which is proportional to the time since the snow started. The answer (an exercise below) involves the golden ratio: it started snowing about 37 minutes before noon. Versions with two ploughs, like this question, followed.

Information that travels

The second plough's motion depends on the history recorded in the road: the time at which each point was last cleared. This is a simple example of a system with *memory*, in which the state at one place depends on what happened there earlier. Traffic flow, the motion of glaciers and the spreading of a forest fire all have this character, and their mathematics, the theory of first-order partial differential equations and their *characteristics*, is a standard second-year topic. The idea of writing time as a function of position ("when does the front arrive here?") is exactly the method of *arrival times* used for such problems.

Try this next

1 **Agnew's problem.** Snow starts falling at a steady rate at an unknown time before noon. A snowplough whose speed is inversely proportional to the depth starts at noon and goes 2 miles in the first hour and 1 mile in the second. Show that if it started snowing t_0 hours before noon, then $(1 + \frac{2}{t_0})^2 = (1 + \frac{1}{t_0})^3$, deduce that $\frac{1}{t_0}$ is

the golden ratio, and find when it started snowing.

2 **Verification.** Verify by differentiation that $t_2(y)$ satisfies the differential equation of part (ii).

3 **The time gap.** Show that if $kT > h$, the time gap $t_2(y) - t_1(y)$ first increases and then decreases to zero, and find where it is greatest. What happens if $kT \leq h$? Interpret this.

4 **Three ploughs.** A third plough starts at time $2T$. Write down the equation for its motion. Does it catch the second plough before or after the second catches the first?

Further reading

- **Ralph Palmer Agnew, *Differential Equations*** (McGraw-Hill, 1942). The original snowplough problem.
- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press), which discusses the snowplough problem among many others.
- **Richard Haberman, *Mathematical Models*** (SIAM). Traffic flow and characteristics, at an accessible level.

The full guided solution is in Section 11.13:
https://beyondstep.keithmcnulty.org/ch11/snowplough.html